WORKED SOLUTION

STEP II 1991, Question 9

Pure • Matrix Groups

STEP II 1991, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q9

The key idea

Everything rests on one fact: the integers modulo 5 form a field. Every non-zero residue has a multiplicative inverse, and a product of non-zero residues is never zero. That is what makes the diagonal entries behave, so closure and inverses come almost for free. For the last part, find the elements of order 1 or 2 explicitly by solving $M^2 = I$, then test whether multiplying two of them keeps you inside the set.

Solution

Throughout, all arithmetic is modulo 5, and we write -1 for 4 when it is convenient. We write a general element as

$$M(a,b,c) = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \qquad a,c \in \{1,2,3,4\},\ b \in \{0,1,2,3,4\}.$$

G is a group

Closure. Multiplying two elements gives

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} d & e \\ 0 & f \end{pmatrix} = \begin{pmatrix} ad & ae+bf \\ 0 & cf \end{pmatrix}.$$

This is upper triangular with entries modulo 5. Its diagonal entries ad and cf are non-zero: since 5 is prime, if $5 \mid ad$ then $5 \mid a$ or $5 \mid d$, and neither holds. So the product lies in G.

Associativity is given, since matrix multiplication is associative.

Identity. $I = M(1,0,1)$ is in G, and $IM = MI = M$ for every M.

Inverses. Every non-zero residue modulo 5 has an inverse ($1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$,

$4^{-1} = 4$). So, given $M(a,b,c) \in G$, we can form

$$N = \begin{pmatrix} a^{-1} & -a^{-1}bc^{-1} \\ 0 & c^{-1} \end{pmatrix},$$

which lies in G because $a^{-1}, c^{-1} \neq 0$. Using the product formula,

$$MN = \begin{pmatrix} aa^{-1} & -aa^{-1}bc^{-1} + bc^{-1} \\ 0 & cc^{-1} \end{pmatrix} = I, \qquad NM = \begin{pmatrix} a^{-1}a & a^{-1}b - a^{-1}bc^{-1}c \\ 0 & c^{-1}c \end{pmatrix} = I.$$

So every element has an inverse in G, and G is a group.

The order of G

There are 4 choices for a, 5 for b and 4 for c, and different choices give different matrices. So

$$|G| = 4 \times 5 \times 4 = 80.$$

Is G commutative?

No. Take $A = M(1,1,1)$ and $B = M(2,0,1)$. Then

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix},$$

so $AB \neq BA$ and G is not commutative.

Elements of order 1 or 2

An element M has order 1 or 2 exactly when $M^2 = I$. By the product formula,

$$M(a,b,c)^2 = \begin{pmatrix} a^2 & b(a+c) \\ 0 & c^2 \end{pmatrix},$$

so we need $a^2 = 1$, $c^2 = 1$ and $b(a+c) = 0$.

The squares of $1, 2, 3, 4$ modulo 5 are $1, 4, 4, 1$, so $a^2 = 1$ forces $a = \pm 1$, and likewise $c = \pm 1$.

- If $a = c$, then $a + c = \pm 2 \neq 0$. Since $\mathbb{Z}_5$ has no zero divisors, $b(a+c) = 0$ forces $b = 0$. This gives I and $-I$.
- If $c = -a$, then $a + c = 0$, so b can be anything. This gives the 10 matrices $M(1, b, -1)$ and $M(-1, b, 1)$, for $b = 0, 1, \ldots, 4$.

So the set H of elements of order 1 or 2 has exactly 12 members.

Ask yourself Is a set of self-inverse elements automatically closed?

No. It contains I, and each element is its own inverse, so the identity and inverse axioms hold automatically. The only thing that can fail is closure, and in a non-commutative

group it usually does: if $X^2 = Y^2 = I$, then $(XY)^2 = XYXY$, and this need not collapse to I unless X and Y commute.

Here it fails. Take $X = M(1,1,-1)$ and $Y = M(1,0,-1)$, both in H. Then

$$XY = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

This has $a = c = 1$ but $b = -1 \neq 0$, so by the classification above it is not in H. (In fact $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}^k = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix}$, so it has order 5.) Hence H is not closed, and it is *not* a subgroup of G.

If you know Lagrange's theorem, there is a quicker argument: the order of a subgroup must divide the order of the group, and $12 \nmid 80$.

Checks

A short computer enumeration of all 80 matrices confirms closure, the inverse formula and $AB \neq BA$ for the pair above. Squaring every element finds exactly the 12 predicted solutions of $M^2 = I$, and $XY = M(1,4,1)$ has order 5. By hand, check that $M(-1,3,1)^2 = I$: the off-diagonal entry is $3(-1+1) = 0$.

Common slips

- Forgetting to show that the inverse lies in G. Its diagonal entries a^{-1}, c^{-1} must be non-zero.
- Missing the elements $M(1,b,-1)$ and $M(-1,b,1)$ with $b \neq 0$ by assuming that $M^2 = I$ forces M to be diagonal.
- Checking only the identity and inverse axioms for H. Closure is the axiom that fails.

Beyond STEP

G is the group of invertible upper triangular 2×2 matrices over the finite field $\mathbb{F}_5$, a *Borel subgroup* of $\mathrm{GL}_2(\mathbb{F}_5)$. The matrices with $a = c = 1$ form a normal subgroup of order 5, and G is built from it and the diagonal matrices as a semidirect product. Groups of matrices over finite fields are a rich source of examples in a first course on group theory. For more, see Allenby, *Rings, Fields and Groups*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.